HOMEWORK 4

1. For a group G define a series of subgroups G_i, $i \geq 0$, by $G_0 = G$ and $G_{i+1} = [G_i, G]$. Prove that G is nilpotent if and only if $G_n = 1$ for some n.

2. For a group G define a series of subgroups $G^{(i)}$, $i \geq 0$, by $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. Prove that G is solvable if and only if $G^{(n)} = 1$ for some n.

3. Find a group of the smallest order that is solvable but not nilpotent.

4. Let $\sigma = (123 \cdots n) \in S_n$. Show that the conjugacy class of σ has $(n-1)!$ elements. Prove that the centralizer of σ in S_n is the cyclic subgroup generated by σ.

5. Show that the alternating group A_4 has no subgroups of order 6.

6. (a) Prove that S_n is generated by $(1,2)$, $(1,3), \ldots (1,n)$.
(b) Prove that S_n is generated by the two cycles $(1,2)$ and $(1,2,\ldots,n)$.

7. (a) Show that the centralizer of A_n in S_n (the subgroup in S_n consisting of all elements, which commute with all elements in A_n) is trivial, if $n \geq 4$.
(b) Let $g \in S_n$ be an odd transformation. Show that the map $f : A_n \to A_n$, given by $f(x) = gxg^{-1}$ is an automorphism. Prove that f is not inner if $n \geq 3$.

8. Prove that every automorphism of S_3 is inner and $Aut(S_3)$ is isomorphic to S_3.

9. Describe all Sylow subgroups in S_n for $n \leq 5$.

10. (a) Show that for any $n \geq 1$, there is an injective homomorphism $S_n \to A_{n+2}$.
(b) Prove that every finite group is isomorphic to a subgroup of a finite simple group.